Arjun Mehra

Robotics / Perception Engineer | Bangalore / Hyderabad / Remote IST | Daytime preferred
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SUMMARY

Robotics engineer with ~2.5 years building perception and ROS2 software for mobile robots. Strong in Python, C++, OpenCV, and embedded (STM32). Experience shipping camera + ranging obstacle avoidance on warehouse AMRs, validated in Gazebo before field trials. Seeking Robotics Engineer roles in perception, control, ROS, or embedded - warehouse, industrial automation, or deep-tech hardware+software teams in India.

EXPERIENCE

Robotics Software Engineer - NovaBot Labs, Bangalore

Mar 2024 - Present

- Owned ROS2 perception nodes for warehouse AMRs: camera + ultrasonic fusion for obstacle avoidance.
- Cut end-to-end perception latency via node graph and QoS changes; validated in Gazebo before field trials.
- Improved aisle-trial collision rate after tuning costmap inflation and safety margins with controls.
- Wrote Python tooling for bag replay and regression checks on perception pipelines.
- Collaborated with mechanical and embedded on sensor mounting, calibration, and bring-up.

Junior Robotics Engineer - MechNest Automation, Bangalore

Aug 2022 - Feb 2024

- Developed ROS (ROS1 to early ROS2) packages for conveyor-adjacent mobile bases.
- Implemented OpenCV line/marker detection for docking assist; C++ nodes for tighter loops.
- Brought up STM32 firmware for motor drivers and basic telemetry over UART/CAN.
- Supported on-site commissioning for two factory pilot deployments (daytime IST).

Embedded Systems Intern - MechNest Automation, Bangalore

Jan 2022 - Jul 2022

- Prototyped sensor drivers (IMU, ultrasonic) and logging utilities on STM32 / Arduino-class boards.
- Assisted senior engineers with hardware bring-up checklists and lab test scripts.

SKILLS

- Robotics middleware: ROS2 (nodes, launch, tf, bags), Gazebo simulation
- Perception: OpenCV, camera calibration basics, obstacle detection, basic SLAM exposure
- Languages: Python, C++, C (embedded)
- Embedded: STM32, UART/CAN basics, motor driver bring-up
- Controls / planning: PID tuning support, costmaps & planners (deepening motion planning)
- Tools: Git, Linux, basic CI for package builds
- Preferences: Bangalore, Hyderabad, or remote IST daytime; lab/field OK; no night-only roles

EDUCATION

B.Tech, Electronics & Communication - RV College of Engineering, Bangalore

2018 - 2022 | Coursework: control systems, embedded, computer vision; ROS line-follower project

PROJECTS

- Gazebo sim stack for AMR aisle navigation demos (costmap + simple planner configs)
- STM32 + ROS bridge prototype for wheel odometry telemetry

This document is a fictional sample CV for product demos. All names and employers are made up.